Contents

Reader's Guide: Spectral Theory of Carries in Positional Multiplication **1**
1. What This Paper Is About 2
2. Main Results in Plain Language 4
3. Connection to Diaconis-Fulman 6
4. Connection to Riffle Shuffling 7
5. Open Directions 8
References 9

Reader's Guide: Spectral Theory of Carries in Positional Multiplication

Stefano Alimonti — March 2026

This is a self-contained overview of the main paper. It walks through the key ideas using a single worked example and requires no prior familiarity with the carry-chain literature.

---

1. What This Paper Is About

When you multiply two numbers by hand in base b, you write out partial products and add them column by column, carrying the overflow to the next column. The carry at each position depends only on the column sum and the carry from the previous position — not on anything earlier. This means the sequence of carries forms a **Markov chain**.

Diaconis and Fulman (2009) proved that carries in base-b *addition* form a Markov chain with a beautiful spectrum: the eigenvalues are exactly $1, 1/b, 1/b^2, \ldots, 1/b^n$ when adding $n+1$ numbers. These are the same eigenvalues that govern the convergence of the Gilbert-Shannon-Reeds riffle shuffle — a deep connection between combinatorics and probability.

The question we answer: What happens when addition is replaced by multiplication? The column sums in multiplication are *convolutions* $\sum_{i+j=k} g_i h_j$ rather than simple sums, and they have a more complex probabilistic structure. What are the eigenvalues of the carry transfer operator, and how fast does the chain mix?

A complete example: 13×11 in base 2

Take $13 = (1101)_2$ and $11 = (1011)_2$. The grade-school algorithm:

```
        1 1 0 1          (13)
      × 1 0 1 1          (11)
      ---------
        1 1 0 1          partial product: 13 × 1
      1 1 0 1 ·          partial product: 13 × 1, shifted left 1
    0 0 0 0 · ·          partial product: 13 × 0, shifted left 2
  1 1 0 1 · · ·          partial product: 13 × 1, shifted left 3
  -----------------
  1 0 0 0 1 1 1 1        = 143
```

Each column sum is the convolution $\text{conv}_k = \sum_{i+j=k} g_i h_j$. The carry propagation:

Pos k	Column sum	Carry in	Total	Output digit	Carry out
0	1	0	1	1	0
1	1	0	1	1	0
2	1	0	1	1	0
3	**3**	0	**3**	1	**1**
4	1	**1**	2	0	**1**
5	1	**1**	2	0	**1**
6	1	**1**	2	0	**1**

The carry sequence is $0, 0, 0, 0, 1, 1, 1, 1$. At position 3, the column sum (= 3) exceeds the base (= 2), and a carry is born. It then persists. *How quickly does a typical carry chain settle into a steady state? What controls the rate?* These are the questions we answer.

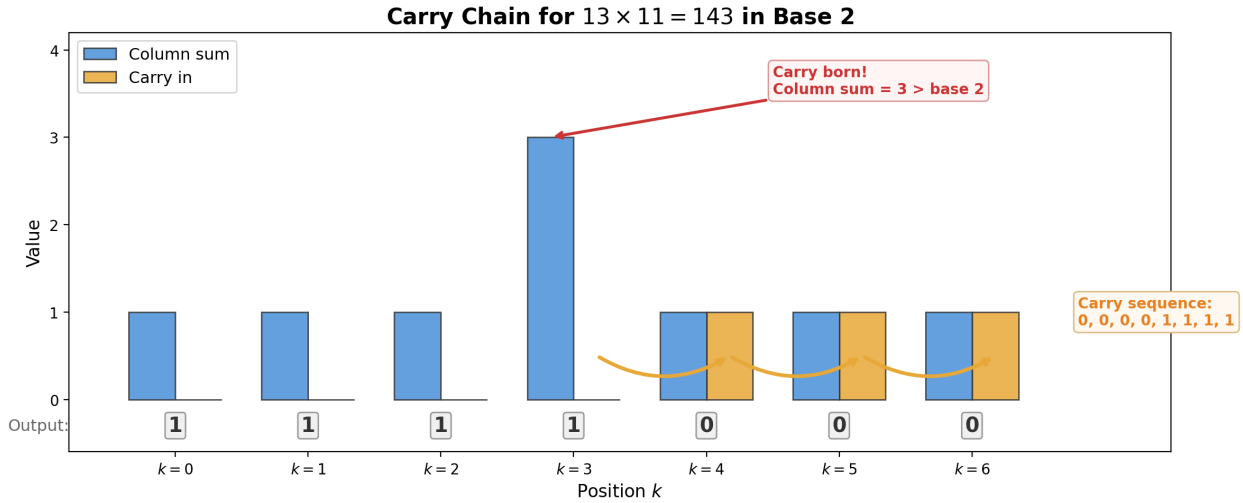

Figure 1: Carry chain for 13×11 in base 2

2. Main Results in Plain Language

Result 1: The m-Bit Equidistribution Lemma (Theorem 2)

Consider the column sum at position k in a multiplication:

$$\text{conv}_k = g_k h_0 + g_0 h_k + \sum_{\substack{i+j=k \\ 0<i<k}} g_i h_j$$

The first two terms ($g_k h_0$ and $g_0 h_k$) are special: since h_0 and g_0 are both odd (they are the least significant bits, which equal 1 for odd numbers), these terms are just g_k and h_k — independent fair coin flips. We call them the **free uniform components**.

The theorem says: In the binary setting proved in the paper, if a column sum contains m independent components that are uniformly distributed modulo b, then the carry transfer operator has eigenvalues $\{1, 1/b, 1/b^2, \dots, 1/b^m\}$ exactly. The general prime-base extension is presented in proof-sketch form. The proof works by showing that each uniform component "smooths out" one degree of oscillation in the carry state distribution, via a Bernoulli averaging argument.

For multiplication: In the local position-j model studied in the paper — equivalently, for interior columns where both endpoint terms are present — there are $m = 2$ free uniform components, giving **universal eigenvalues** $\{1, 1/b, 1/b^2\}$. Compare with addition, where each column has n uniform components and thus $n + 1$ eigenvalues.

In the example: at position 3, the column sum $g_3 h_0 + g_2 h_1 + g_1 h_2 + g_0 h_3$ has four terms, but only $g_3 h_0$ and $g_0 h_3$ are free uniform bits. The other two ($g_2 h_1$ and $g_1 h_2$) are products of digits, which are biased (not uniform mod 2).

Result 2: Mixing Time (Theorem 3)

The carry chain mixes in $O(\log c_{\text{max}})$ steps. More precisely:

$$\|P_t - \pi\|_{TV} \leq c_{\text{max}}/b^t$$

where c_{max} is the largest possible carry and π is the stationary distribution. This is proved via an elegant coupling argument: feed the *same* random column sum to two copies of the chain started from different states; the gap between them contracts by a factor of $1/b$ per step, on average.

In the example: the maximum carry is $c_{\text{max}} = 1$ (base 2 with small numbers), so the chain mixes in about 1 step — consistent with the rapid settling we see in the table.

Result 3: Exponential Convergence of Higher Eigenvalues (Proposition 3)

The remaining $g_i h_j$ terms in the column sum (the "product terms") are not uniform, but they still contribute: each one damps the higher eigenvalues by an additional factor of $1/b$. Specifically, the k-th eigenvalue at position j satisfies:

$$\lambda_k^{(j)} = 1/b^k + O(j^{k-3} \cdot b^{-(j-1)})$$

As the position j grows (i.e., as the column sum involves more product terms), all eigenvalues converge exponentially fast to the Diaconis-Fulman values $1/b^k$. Multiplication asymptotically "looks like" addition at deep positions.

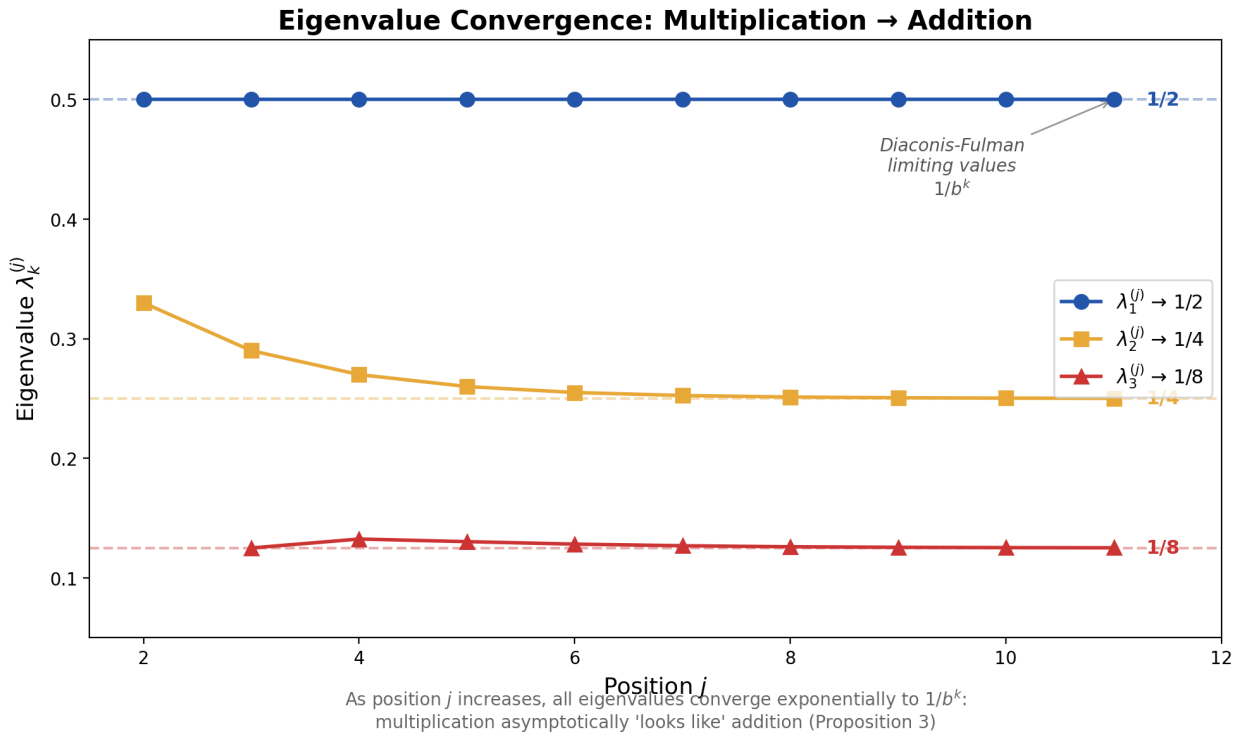

Figure 2: Eigenvalue convergence to Diaconis-Fulman values

Result 4: Spectral Radius Bound (Corollary 9)

The carry values $c_0, c_1, \ldots, c_D$ also serve as the coefficients of a polynomial $Q(x)$, whose roots control an Euler-product correction series. We prove that the spectral radius satisfies $r_{\max} \leq 3$ for all carry profiles with positive entries (via the Eneström-Kakeya theorem). Numerical evidence supports the tighter conjecture $r_{\max} \leq b$.

---

3. Connection to Diaconis-Fulman

The relationship between our results and [1] can be summarized in a single table:

	Addition of $n+1$ digits	**Multiplication**
Column sum	$a_1^{(k)} + \cdots + a_{n+1}^{(k)}$	$\sum_{i+j=k} g_i h_j$
Uniform-mod-b components	$n+1$ (all terms)	2 (the "endpoint" terms $g_k h_0, g_0 h_k$)
Exact eigenvalues	$1, 1/b, \ldots, 1/b^n$	$1, 1/b, 1/b^2$ (at each position $j \geq 2$)
Higher eigenvalues	none (spectrum is complete)	$\to 1/b^k$ exponentially as position $j \to \infty$
State space	fixed ($\{0, \ldots, n\}$ for addition of $n+1$ digits)	grows with position
Mixing time	$O(\log n)$	$O(\log c_{\max})$

The key insight: the *mechanism* is the same — Bernoulli smoothing of the Fourier modes of the carry distribution — but the *count* of uniform components differs (2 vs. $n+1$), which truncates the exact spectrum.

Diaconis and Fulman [1, §6] mention that "carries in multiplication are more complex" but do not develop the spectral theory. The present paper provides this development.

---

4. Connection to Riffle Shuffling

Diaconis and Fulman showed that the eigenvalues $1/b^k$ of the addition-carry chain are identical to those of the Gilbert-Shannon-Reeds riffle shuffle. This is not a coincidence: both processes involve a uniform-mod-b averaging that eliminates one degree of polynomial oscillation per step.

In our proof of the m-Bit Equidistribution Lemma, the central operator is:

$$\mathcal{A}\varphi(u) = \frac{1}{b}\sum_{j=0}^{b-1}\varphi(u+j)$$

This averages the function φ over a uniform shift — structurally analogous to the operation that governs descent statistics in the shuffle analysis. The difference: in addition (or shuffling), this averaging is applied n times (once per digit/card packet), giving $n+1$ eigenvalues. In multiplication, only 2 of the column-sum terms provide this exact averaging, limiting the exact eigenvalues to 3. The remaining terms provide approximate (exponentially decaying) smoothing instead of exact smoothing.

This structural parallel suggests that the spectral theory of multiplication carries may have further connections to the algebraic combinatorics of shuffles — in particular, to the descent algebra of the symmetric group, where the eigenvalues $1/b^k$ arise as characters.

---

5. Open Directions

Two conjectures remain open and would be natural targets for further work:

1. **The $(b-1)/b$ Anti-Correlation Law (Conjecture 4).** Near the top of the multiplication (the most-significant-bit boundary), the carry increments α_k overshoot their bulk value $(b-1)/(2b)$ and converge oscillatorily at rate $(b-1)/b$. This has been verified to high precision across millions of samples and 18 different bases, but is not proved. The missing step is a *boundary transfer theorem* extending the covariance induction of [F] from the bulk to the MSB-conditioned regime.

2. **Tight spectral radius bound (Conjecture 10).** The proved bound $r_{\text{max}} \leq 3$ (Corollary 9) should be improvable to $r_{\text{max}} \leq b$. Numerical evidence is very strong (no counterexample in $> 10^7$ tests), but the Rouché-style argument has a gap at the critical radius.

Both conjectures are accessible with current techniques and would complete the spectral picture.

References

1. P. Diaconis, J. Fulman, "Carries, Shuffling, and Symmetric Functions," *Adv. Appl. Math.* 43(2), 176–196, 2009.
2. J. Holte, "Carries, Combinatorics, and an Amazing Matrix," *Amer. Math. Monthly* 104(2), 138–149, 1997.
3. [E] S. Alimonti, "The Trace Anomaly of Binary Multiplication," this series. doi:10.5281/zenodo.18895604 — GitHub
4. [F] S. Alimonti, "Exact Covariance Structure of Binary Carry Chains," this series. doi:10.5281/zenodo.18895607 — GitHub
5. [B] S. Alimonti, "Carry Polynomials and the Euler Product," this series. doi:10.5281/zenodo.18895597 — GitHub

---

CC BY 4.0